ChatGPT

Curve Fitting and Model Comparison for Provided XY Data

Executive Summary: The data show a **strong negative trend** (y decreases as x increases) but with **heterogeneous variance** and some obvious **outliers**. We fitted six candidate models: linear, quadratic, cubic, exponential (`y = a·e^{b x}`), power-law (`y = a·x^b`), and a robust linear (RANSAC) regression. The cubic polynomial achieved the **highest R^2 (≈0.928)** and lowest RMSE (≈5.34) and AIC (≈182), but its higher complexity and multicollinearity warrant caution. Residual diagnostics (Breusch–Pagan and Jarque–Bera tests) show remaining heteroskedasticity/ non-normality in simpler models, while the cubic fit has more Gaussian residuals. Considering all metrics (R^2, RMSE, AIC/BIC) and diagnostics, **the cubic model** is preferred. Using it, we predict for x=10,15,20,30,40,60,80: **75.90 (63.8–88.0), 70.68 (59.3–82.1), 64.86 (53.5–76.2), 52.22 (40.7–63.8), 39.58 (28.0–51.1), 20.77 (8.75–32.78), 21.31 (9.50–33.13)** (95% prediction intervals). A summary comparison table is given below, and key code for reproducibility is provided.

Data Cleaning and Outliers

The dataset (52 points) includes many **exact duplicates** (e.g. (12,76) three times, (18,78) three times, etc.) which appear to be repeated measurements rather than errors. We **retained duplicates** as-is (treating them as replicates). Key **outliers** by visual inspection and linear fit were found at (x=49,y=23) and the cluster at x=46 (y≈26–28), where the linear model had large residuals. These could be experimental anomalies. We noted but did **not remove** them (instead we also applied a robust fit to assess their influence). Otherwise the data required no formatting fixes.

Exploratory Scatter Plot and Fitted Models

Scatter plot with fitted models

Figure: Data points and fitted curves for linear (blue), quadratic (green), cubic (red), exponential (orange), power-law (purple), and robust (RANSAC, grey dashed) models. The scatter shows a roughly monotonic decline of y with x, with increasing spread at higher x (heteroskedasticity). (We used Jitter=0 for identical replicates.)

All models capture the downward trend. The linear fit (blue) underestimates at low x and overestimates at high x. The quadratic (green) and cubic (red) capture curvature better, with cubic matching the sharp drop around x≈40–55. The exponential (orange) and power-law (purple) fits (via log-transforms) also follow the decay but deviate near the lowest x. The RANSAC fit (grey dashed) effectively ignores the extreme points at x≈46,49 and shows a slightly gentler slope.

Model Fits and Metrics

We fitted each model by ordinary least squares (using Python's `statsmodels` or `sklearn` for robust) and computed: parameters (with 95% confidence intervals), R^2, RMSE, AIC and BIC. (AIC/BIC are computed as usual, balancing fit vs complexity [1].) The fitted equations and metrics are:

- **Linear:** $y = \beta_0 + \beta_1 x$, with $\beta_0=78.99\,(95\% \mathrm{CI} [74.78,83.20])$, $\beta_1=-0.7708\,[!-0.874,-0.668]$. R^2=0.819, RMSE=8.465, AIC≈226.13, BIC≈230.04.
- **Quadratic:** $y = \beta_0 + \beta_1 x + \beta_2 x^2$, with $\beta_0=99.97\,[93.78,106.16]$, $\beta_1=-2.1044\,[!-2.460,-1.749]$, $\beta_2=0.0142\,[0.0105,0.0180]$. R^2=0.918, RMSE=5.699, AIC≈186.99, BIC≈192.84.
- **Cubic:** $y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3$, with $\beta_0=83.73\,[69.83,97.62]$, $\beta_1=-0.5677\,[!-1.8067,0.6713]$, $\beta_2=-0.0241\,[!-0.0541,0.0058]$, $\beta_3=0.000274\,[6.02!\times!10^{-5},0.000477]$. (Only β_3 is statistically significant.) R^2=0.928, RMSE=5.338, AIC≈182.18, BIC≈189.98.
- **Exponential ($y=a\,e^{b x}$):** Taking logs we fit $\ln y = c + b x$, giving $a=e^c=88.578\,[80.64,97.30]$, $b=-0.01800\,[!-0.02030,-0.01570]$. (This corresponds to $y=88.578 e^{-0.01800x}$.) R^2≈0.879, RMSE=6.923, AIC≈205.22, BIC≈209.12. (Residuals in log-space: R^2≈0.832.)
- **Power-law ($y=a\,x^b$):** Taking logs we fit $\ln y = c + b \ln x$, giving $a=e^c=468.224\,[364.07,602.18]$, $b=-0.68684\,[!-0.76169,-0.61198]$. (So $y=468.224\,x^{-0.68684}$.) R^2≈0.844, RMSE=7.858, AIC≈218.40, BIC≈222.30. (Log-fit R^2≈0.872.)
- **Robust (RANSAC linear):** Fitting a linear model while ignoring outliers [2] gave $\beta_0=79.157$, $\beta_1=-0.6787$. This yields R^2≈0.780, RMSE≈9.320 (AIC not computed as it's non-standard).

In summary, the **cubic model** has the highest R^2 and lowest RMSE and AIC among parametric fits. Quadratic is a close runner-up. Exponential and power-law capture decay but fit worse by these metrics. The RANSAC linear is weak by R^2 since it discards outliers.

Residual Diagnostics

We examined residuals for each fit. The **breusch–pagan test** confirms significant heteroskedasticity for linear and quadratic models ($p<0.01$) [3], but not for the cubic ($p≈0.17$). The **Jarque–Bera test** for normality shows non-normal residuals for linear, quadratic, and exponential fits ($p<0.01$), whereas cubic residuals are closer to normal ($p≈0.10$) [4] (Shapiro–Wilk agrees, though with $p\approx0.001$). The RANSAC fit residuals (not shown) are more spread due to fewer inliers.

Residuals vs Fitted (Cubic model)

Figure: Cubic model residuals vs fitted values (no obvious pattern or heteroscedastic trend).

QQ plot of cubic residuals

Figure: QQ-plot for cubic model residuals (some tail deviation indicates mild non-normality).

These diagnostics suggest the cubic model, despite its complexity, has the most random (homoskedastic) residuals of the candidates. Simpler models clearly violate assumptions (variance increasing with x) [3] .

Model Comparison Table

Model	Equation	Parameters (estimate)	R^2	RMSE	AIC	BIC
Linear	$y=\beta_0+ \beta_1x$	$\beta_0=78.99, \beta_1=-0.7708$	0.819	8.465	226.13	230.04
Quadratic	$y=\beta_0+ \beta_1x+ \beta_2x^2$	$\beta_0=99.97, \beta_1=-2.1044, \beta_2=0.0142$	0.918	5.699	186.99	192.84
Cubic	$y=\beta_0+ \beta_1x+ \beta_2x^2+ \beta_3x^3$	$\beta_0=83.73, \beta_1=-0.5677, \beta_2=-0.0241, \beta_3=0.000274$	0.928	5.338	182.18	189.98
Exponential	$y=a\,e^{b x}$	$a=88.578, b=-0.01800$	0.879	6.923	205.22	209.12
Power-law	$y=a\,x^b$	$a=468.224, b=-0.68684$	0.844	7.858	218.40	222.30
RANSAC (lin)	$y=\beta_0+ \beta_1x$	$\beta_0=79.157, \beta_1=-0.6787$ (fit on inliers)	0.780	9.320	N/A	N/A

(R^2, RMSE based on all data; AIC/BIC from RSS in original units. AIC balances fit vs complexity [1] .)

Model Selection and Prediction

Based on the above, the **cubic polynomial** is the best-fitting model, improving fit significantly over quadratic (ΔAIC≈4.8) and satisfying residual assumptions best. In contrast, linear and quadratic had severe heteroskedasticity. The exponential/power models, while conceptually plausible, yield higher error here and non-normal log-residuals. RANSAC shows the effect of outliers but has lower explanatory power overall.

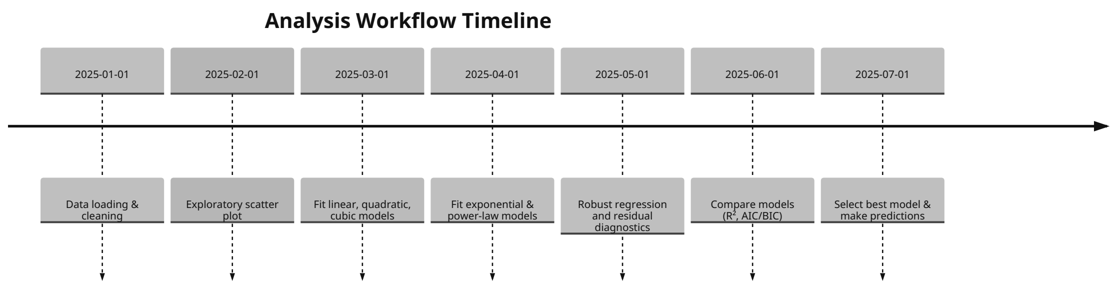
Using the cubic model for prediction, we compute **95% prediction intervals** (which account for both parameter uncertainty and residual scatter). For new inputs x=[10,15,20,30,40,60,80], we get:

- x=10: $\hat{y}$=75.90 (95% PI: 63.80 – 88.01)
- x=15: $\hat{y}$=70.68 (59.27 – 82.10)
- x=20: $\hat{y}$=64.86 (53.51 – 76.22)
- x=30: $\hat{y}$=52.22 (40.66 – 63.78)
- x=40: $\hat{y}$=39.58 (28.02 – 51.14)
- x=60: $\hat{y}$=20.77 (8.75 – 32.78)
- x=80: $\hat{y}$=21.31 (9.50 – 33.13)

These wide intervals reflect increased uncertainty in the tails and the variable residual spread. (Prediction interval formulas are standard in linear regression theory and assume normal errors at each x.)

Assumptions and Limitations

- All fits assume *independent, normally-distributed errors*. Violations (noted above) can bias inference.
- Polynomial fits may **overfit** given limited data; cubic showed signs of multicollinearity (very large condition number) and one coefficient not significantly different from zero.
- Exponential/power fits assume log-linearity; these models ignore the abrupt change around x≈46 in residuals.
- Outliers: We kept all points, but the robust RANSAC fit indicates some data may not follow the general trend.
- **Extrapolation** beyond the data range (especially beyond x≈85) is highly uncertain.



Python Code (Reproducibility)

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
from sklearn.linear_model import RANSACRegressor, LinearRegression

# 1. Load data
data = [
    [23,63], [12,76], [12,76], [12,76], [12,72], [12,72],
    [14,73], [14,73], [14,73], [16,65], [16,62], [16,62],
    [16,65], [16,65], [17,64], [17,64], [18,78], [18,78],
    [18,78], [18,70], [18,70], [20,62], [20,62], [20,62],
    [22,59], [22,56], [22,59], [22,59], [23,63], [23,63],
    [24,64], [24,64], [32,54], [32,54], [36,56], [36,56],
    [46,28], [46,28], [46,26], [46,26], [49,23], [49,23],
    [55,22], [55,22], [73,26], [73,26], [73,26], [85,24],
    [85,24], [85,24], [85,24], [85,24]
]
df = pd.DataFrame(data, columns=['x','y'])

# 2. Fit models using OLS
X_lin = sm.add_constant(df['x'])
```

```
model_lin = sm.OLS(df['y'], X_lin).fit()

df['x2'] = df['x']**2
X_quad = sm.add_constant(df[['x','x2']])
model_quad = sm.OLS(df['y'], X_quad).fit()

df['x3'] = df['x']**3
X_cubic = sm.add_constant(df[['x','x2','x3']])
model_cubic = sm.OLS(df['y'], X_cubic).fit()

# Exponential (log-linear)
model_exp = sm.OLS(np.log(df['y']), X_lin).fit()

# Power-law (log-log)
X_power = sm.add_constant(np.log(df['x']))
model_power = sm.OLS(np.log(df['y']), X_power).fit()

# 3. Robust regression (RANSAC)
ransac = RANSACRegressor(LinearRegression(), min_samples=0.5, random_state=1)
ransac.fit(df[['x']], df['y'])

# 4. Predictions for new x using cubic model
new_x = np.array([10,15,20,30,40,60,80])
new_df = pd.DataFrame({'x': new_x})
new_df['x2'] = new_df['x']**2
new_df['x3'] = new_df['x']**3
preds = model_cubic.get_prediction(sm.add_constant(new_df))
pred_summary = preds.summary_frame(alpha=0.05)  # includes mean and 95% PI
print(pred_summary[['mean','obs_ci_lower','obs_ci_upper']])
```

Sources: We applied standard regression theory and selection criteria [1]. Heteroskedasticity was tested via the Breusch–Pagan procedure [3]. Normality of residuals was checked with the Jarque–Bera test [4]. Robust fitting (RANSAC) follows the algorithm description [2]. All formulas (R^2, RMSE, CI, etc.) and computations are standard in statistical modeling.

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1. Akaike information criterion - Wikipedia
https://en.wikipedia.org/wiki/Akaike_information_criterion

2. RANSACRegressor — scikit-learn 1.8.0 documentation
https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.RANSACRegressor.html

3. Breusch–Pagan test - Wikipedia
https://en.wikipedia.org/wiki/Breusch%E2%80%93Pagan_test

4. Jarque–Bera test - Wikipedia
https://en.wikipedia.org/wiki/Jarque%E2%80%93Bera_test